28 de Julio de 2014

1)

$$y[n-1] - \frac{5}{2}y[n] + y[n+1] = x[n]$$

TRANSFORTO:

$$z^{-1}Y(z) - \frac{5}{2}Y(z) + zY(z) = X(z)$$

$$Y(z)\left\{z - \frac{5}{2} + z^{-1}\right\} = X(z)$$

$$Y(z)\left\{z^2 - \frac{5}{2}z + 1\right\} = zX(z)$$

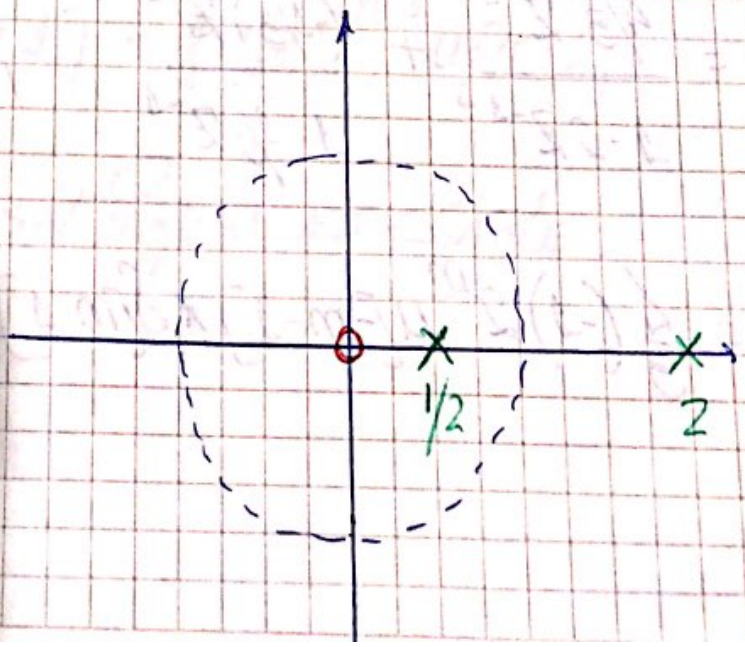
$$\frac{Y(z)}{X(z)} = H(z) = \frac{z}{z^2 - 5/2\, z + 1}$$

$$\frac{5/2 \pm 3/2}{2} \begin{cases} \nearrow 2 \\ \searrow 1/2 \end{cases}$$

$$H(z) = \frac{z}{(z-2)(z-1/2)}$$

CERO: $z = 0$

POLOS: $z = 2 \wedge z = 1/2$



Scanned with CamScanner

3 POSIBLES SOLUCIONES: a) $|Z| < 1/2$.

b) $1/2 < |Z| < 2$

c) $|Z| > 2$.

a) EL SISTEMA NO ES ESTABLE NI CAUSAL

$$H(z) = \frac{Z}{(Z-2)(Z-1/2)} = \frac{A}{(Z-2)} + \frac{B}{(Z-1/2)}$$

$$A = (z-2)\,H(z)\Big|_{z=2} = \frac{2}{(2-1/2)} = 4/3$$

$$B = (Z-1/2)\,H(z)\Big|_{Z=1/2} = \frac{1/2}{1/2-2} = -1/3$$

$$H(z) = \frac{4/3}{z-2} + \frac{(-1/3)}{z-1/2} \quad ; \quad |Z| < 1/2.$$

$$H(z) = \frac{4/3\, z^{-1}}{1-2z^{-1}} + \frac{(-1/3)z^{-1}}{1-1/2\, z^{-1}} \quad ; \quad |Z| < 1/2$$

$$h[m] = \frac{4}{3}(-1)2^m u[-m-1] * \delta[m-1] + \left(-\frac{1}{3}\right)(-1)\left(\frac{1}{2}\right)^m u[-m-1] * \delta[m-1]$$

Scanned with CamScanner

$$h[m] = -\frac{4}{3} 2^{m-1} u[-m] + \frac{1}{3} \left(\frac{1}{2}\right)^{m-1} u[-m]$$

b) $$H(z) = \frac{4/3\, z^{-1}}{1-2z^{-1}} + \frac{(-1/3) z^{-1}}{1-1/2\, z^{-1}}, \quad 1/2 < |z| < 2.$$

$$h[m] = -\frac{4}{3} 2^{m-1} u[-m] + \left(\frac{-1}{3}\right)\left(\frac{1}{2}\right)^{m} u[m] * \delta[m-1].$$

$$h[m] = -\frac{4}{3} 2^{m-1} u[-m] - \frac{1}{3} \left(\frac{1}{2}\right)^{m-1} u[m-1]$$

↳ EL SISTEMA ES ESTABLE PERO NO ES CAUSAL.

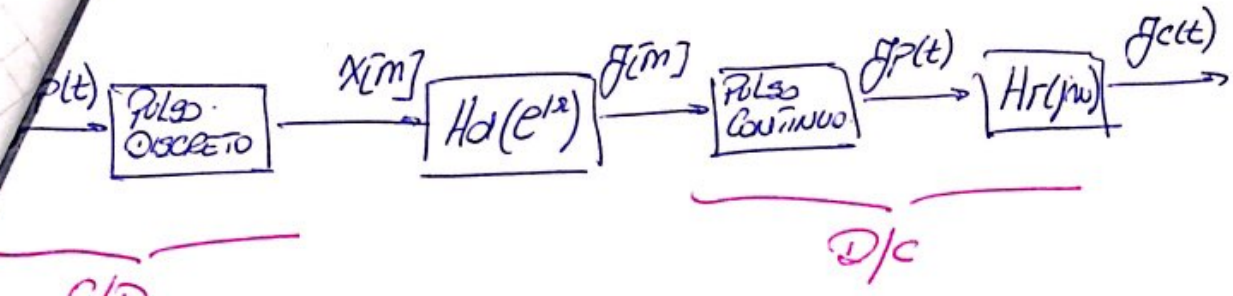
c) $$H(z) = \frac{4/3\, z^{-1}}{1-2z^{-1}} + \frac{(-1/3) z^{-1}}{1-1/2\, z^{-1}}, \quad |z| > 2$$

$$h[m] = \frac{4}{3} 2^{m} u[m] * \delta[m-1] - \frac{1}{3} \left(\frac{1}{2}\right)^{m} u[m] * \delta[m-1]$$

$$h[m] = \frac{4}{3} 2^{m-1} u[m-1] - \frac{1}{3} \left(\frac{1}{2}\right)^{m-1} u[m-1]$$

↳ EL SISTEMA ES CAUSAL PERO NO ES ESTABLE

Scanned with CamScanner



$$Sc(t) = Xc(t) + \beta\, Xc(t-z) \quad ; \quad |\beta| < 1 \quad ; \quad z > 0.$$

$$Hr(jw) = \begin{cases} T & ; |w| \le \pi/T \\ 0 & ; \text{otro caso.} \end{cases} \quad ; \quad T \text{ PERIODO DE MUESTREO.}$$

a) Busco que $gc(t) = Xc(t)$

$$Yc(jw) = \frac{1}{T} Hd(e^{jwT})\, Hr(jw)\, Xc(jw) \; ; \; w \in \mathbb{R}$$

$$Hc(jw) = \frac{1}{T} Hd(e^{jwT})\, Hr(jw)$$

$$\rightarrow Hc(jw) = \begin{cases} Hd(e^{jwT}) & ; |w| \le \pi/T \\ 0 & ; |w| > \pi/T \end{cases}$$

$$Sc(jw) = Xc(jw)\left(1 + \beta e^{-jwz}\right) \qquad \begin{cases} Hc(jw) = \dfrac{1}{1+\beta e^{-jwz}} \\ Yc(jw) = Xc(jw) \end{cases}$$

SIENDO $wT = \Omega \Rightarrow w = \Omega/T$

$$\Rightarrow Hd(e^{j\Omega}) = \frac{1}{1 + \beta e^{-j\Omega z/T}} \; ; \; |\Omega| \le \pi$$

Scanned with CamScanner

$\omega_s \geq 2W \Rightarrow \frac{2\pi}{T} \geq 2W \Rightarrow T \leq \pi/W$

PARA: $\boxed{T = \pi/W}$

↳ $\boxed{H_d(e^{j\Omega}) = \frac{1}{1+\beta e^{-j\frac{\Omega}{\pi} W}}}$

b) $H_d(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1}{1+\beta e^{-j\frac{\Omega}{\pi} W}}$

$\Rightarrow Y(e^{j\Omega})\left(1+\beta e^{-j\frac{\Omega}{\pi} W}\right) = X(e^{j\Omega})$

$\Rightarrow \boxed{y[m] + \beta y[m - \frac{W}{\pi}] = x[m]}$

↳ NECESARIAMENTE: $\frac{W}{\pi} \in \mathbb{Z}$

Scanned with CamScanner

3) $x[m]$ DE LONGITUD FINITA.

$X[k]$ DFT DE N PUNTOS DE $x[m]$

$$x_1[m] = x[m]\cos\left(\frac{2\pi m\ell}{N}\right) \; ; \; m=0,\dots,N-1$$

$$x_2[m] = x[m]\,\text{SEN}\left(\frac{2\pi m\ell}{N}\right) \; ; \; m=0,\dots,N-1$$

DONDE $\ell = 0,\dots,N-1$

$$e^{j\frac{2\pi m\ell}{N}} = \cos\left(\frac{2\pi m\ell}{N}\right) + j\,\text{SEN}\left(\frac{2\pi m\ell}{N}\right)$$

$$x[m]\,e^{j\frac{2\pi m\ell}{N}} = x[m]\cos\left(\frac{2\pi m\ell}{N}\right) + j\,x[m]\,\text{SEN}\left(\frac{2\pi m\ell}{N}\right)$$

$$x[m]\,e^{j\frac{2\pi m\ell}{N}} = x_1[m] + j\,x_2[m]$$

(1) $x_1[m] + j x_2[m] = x[m]\,e^{j\frac{2\pi m\ell}{N}} \rightarrow x[m]\,W_N^{-m\ell}$

(2) $x_1[m] - j x_2[m] = x[m]\,e^{-j\frac{2\pi m\ell}{N}} \rightarrow x[m]\,W_N^{m\ell}$

PROPIEDAD: $W_N^{-m\ell}\,x[m] \rightarrow X[k-\ell]$; $W_N = e^{-j\frac{2\pi}{N}}$

CALCULO LAS DFT's

(1) $X_1[k] + j X_2[k] = X[k-\ell]$

(2) $X_1[k] - j X_2[k] = X[k+\ell]$

$$\Rightarrow X_1[k] = \frac{X[k-\ell] + X[k+\ell]}{2} \; ; \; X_2[k] = \frac{X[k-\ell] - X[k+\ell]}{2j}$$

Scanned with CamScanner